Hafis Muhammed K

Generative AI Engineer

+91 8089305091 hafizmuhammed1019@gmail.com Portfolio hafis-muhammed-k Hafizmuhammedk

Technical Skills

Programming Languages: Python, R, Javascript
Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, roboflow, LiveKit, NLTK, MLflow, DeepEval, Ragas
Backend Framework: FastAPI, Flask
Data & Visualization: Feature Engineering, ggplot2, TensorBoard
MLOps & Deployment: Docker, GitHub Actions (CI/CD), MLflow
Cloud Platforms & Services: AWS (ECR, EC2, ECS, S3, CloudWatch), GCP (VM Instance, Vertex AI)
Database & Vector Store: PostgreSQL, Redis, Valkey, Pinecone, Airtable
Embedded & IoT: ESP32, Arduino IDE, MicroPython, PlatformIO
Tools: Roboflow, Jupyter Notebook

Work Experience

Generative AI Engineer Intern Sep 2025 – Presnt
Mentrex *Malappuram, Kerala*

- Worked on real-time **full-duplex voice pipelines** using **LiveKit**, integrating streaming speech processing for low-latency, continuous voice conversations.
- Implemented **multi-agent orchestration and handoffs** using **Agno**, enabling intelligent routing and seamless conversation flow between specialized AI agents.
- Integrated **Redis caching and state management** into AI pipelines to store conversation data and frequently accessed information, improving response efficiency and pipeline performance.

Projects

DropBook AI 2026
[Python, LiveKit Agents, LiveKit, Deepgram, FastAPI, Twilio, SQLAlchemy, WebRTC]

- Engineered a low latency, full duplex AI voice pipeline for browser and phone conversations using **LiveKit WebRTC/SIP**, with streaming **STT**, LLM reasoning, **TTS**, **VAD**, and interruption handling.
- Designed **FastAPI** services for agent configuration, call management, and synchronized transcripts, backed by **SQLAlchemy** persistence and **Docker/GCP deployment**.

Face Detection & Recognition 2025
[Python, OpenCV, MediaPipe, YOLOv8, ONNX]

- Implemented **YOLOv8** inference for multi person face recognition with confidence based identity detection and compatibility with webcam and **RTSP/IP camera** streams.
- Converted the **YOLOv8n** base model to **ONNX** for optimized inference on **Raspberry Pi 5**, achieving higher **FPS** and faster inference compared to the base model.
- Generated and organized **YOLO** training datasets with automated annotations, class mapping, train/validation splitting, and data validation.

Water Monitoring System
[ESP32, Python, Arduino IDE, Water Flow & Level Sensors]

- Designed an **ESP32 based real time water monitoring system** using flow and level sensors with threshold based event detection and sensor noise filtering.
- Optimized the monitoring loop for continuous sensor sampling and event classification, reducing false positive detections by 12% during testing.

Education

Bachelor of Computer Applications (BCA) 2026 – Present
Indira Gandhi National Open University (IGNOU)

Higher Secondary School 2023 – 2025
St.Paul's EMHSS (Plus Two)